

```
import warnings
warnings.filterwarnings('ignore')
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

df = pd.read_csv('/content/Airbnb_data.csv', skipinitialspace = True )
df.head()
```

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	4
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	4
2	1002403	THE VILLAGE OF HARLEM....NE W YORK !	78829239556	NaN	Elise	Manhattan	Harlem	4
3	1002755	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	4
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	4

5 rows × 26 columns

```
df.columns

Index(['id', 'NAME', 'host id', 'host_identity_verified', 'host name',
      'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
      'country code', 'instant_bookable', 'cancellation_policy', 'room
type',
      'Construction year', 'price', 'service fee', 'minimum nights',
      'number of reviews', 'last review', 'reviews per month',
      'review rate number', 'calculated host listings count',
      'availability 365', 'house_rules', 'license'],
      dtype='object')
```

Checking missing values

```
print(df.isnull().sum())

id                                0
NAME                             250
host id                           0
host_identity_verified            289
host name                         406
neighbourhood group               29
neighbourhood                    16
lat                               8
long                              8
country                           532
country code                      131
instant_bookable                  105
cancellation_policy               76
room type                         0
Construction year                 214
price                             247
service fee                       273
minimum nights                    409
number of reviews                 183
last review                      15893
reviews per month                 15879
review rate number                 326
calculated host listings count    319
availability 365                  448
```

```

house_rules          52131
license              102597
dtype: int64

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 102599 entries, 0 to 102598
Data columns (total 26 columns):
#   Column                                  Non-Null Count  Dtype
---  -
0   id                                       102599 non-null  int64
1   NAME                                    102349 non-null  object
2   host id                                 102599 non-null  int64
3   host_identity_verified                 102310 non-null  object
4   host name                             102193 non-null  object
5   neighbourhood group                   102570 non-null  object
6   neighbourhood                         102583 non-null  object
7   lat                                    102591 non-null  float64
8   long                                   102591 non-null  float64
9   country                               102067 non-null  object
10  country code                          102468 non-null  object
11  instant_bookable                      102494 non-null  object
12  cancellation_policy                   102523 non-null  object
13  room type                             102599 non-null  object
14  Construction year                     102385 non-null  float64
15  price                                  102352 non-null  object
16  service fee                           102326 non-null  object
17  minimum nights                        102190 non-null  float64
18  number of reviews                     102416 non-null  float64
19  last review                           86706 non-null   object
20  reviews per month                     86720 non-null   float64
21  review rate number                    102273 non-null   float64
22  calculated host listings count         102280 non-null   float64
23  availability 365                       102151 non-null   float64
24  house_rules                           50468 non-null   object
25  license                                2 non-null       object
dtypes: float64(9), int64(2), object(15)
memory usage: 20.4+ MB

```

Handling Missing Values

```

df['last review'] = pd.to_datetime(df['last review'], errors = 'coerce')

df['reviews per month'] = df['reviews per month'].fillna(0)

print("Missing last review:", df['last review'].isna().sum())
print("Missing reviews per month:", df['reviews per month'].isna().sum())

Missing last review: 50228
Missing reviews per month: 0

```

```

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 102599 entries, 0 to 102598
Data columns (total 26 columns):
#   Column                                  Non-Null Count  Dtype
---  -
0   id                                       102599 non-null  int64
1   NAME                                    102349 non-null  object
2   host id                                 102599 non-null  int64
3   host_identity_verified                 102310 non-null  object
4   host name                             102193 non-null  object
5   neighbourhood group                   102570 non-null  object
6   neighbourhood                         102583 non-null  object
7   lat                                    102591 non-null  float64
8   long                                   102591 non-null  float64
9   country                               102067 non-null  object
10  country code                          102468 non-null  object
11  instant_bookable                      102494 non-null  object
12  cancellation_policy                   102523 non-null  object
13  room type                             102599 non-null  object
14  Construction year                     102385 non-null  float64
15  price                                  102352 non-null  object
16  service fee                           102326 non-null  object
17  minimum nights                        102190 non-null  float64
18  number of reviews                     102416 non-null  float64
19  last review                           52371 non-null   datetime64[ns]
20  reviews per month                     102599 non-null  float64
21  review rate number                    102273 non-null  float64
22  calculated host listings count         102280 non-null  float64
23  availability 365                       102151 non-null  float64
24  house_rules                           50468 non-null  object
25  license                                2 non-null       object
dtypes: datetime64[ns](1), float64(9), int64(2), object(14)
memory usage: 20.4+ MB

```

```
df.dropna(subset= ['NAME', 'host name'], inplace = True)
```

```
print(df.isnull().sum())
```

```

id          0
NAME        0

```

```

host id                                0
host_identity_verified                 276
host name                             0
neighbourhood group                   26
neighbourhood                         16
lat                                   8
long                                  8
country                               526
country code                          122
instant_bookable                      96
cancellation_policy                   70
room type                             0
Construction year                     200
price                                 239
service fee                           268
minimum nights                        403
number of reviews                     182
last review                           49879
reviews per month                      0
review rate number                    314
calculated host listings count        318
availability 365                      420
house_rules                           51867
license                               101947
dtype: int64

df = df.drop(columns = ['license', 'house_rules'], errors = 'ignore')

df.head()

```

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	4
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	4
2	1002403	THE VILLAGE OF HARLEM....NE W YORK !	78829239556	NaN	Elise	Manhattan	Harlem	4
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	4
5	1004098	Large Cozy 1 BR Apartment In Midtown East	45498551794	verified	Michelle	Manhattan	Murray Hill	4

5 rows × 24 columns

```

df['price'] = df['price'].replace('[\$,]', '', regex = True).astype(float)
df['service fee'] = df['service fee'].replace('[\$,]', '', regex=
True).astype(float)

df.head()

```

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	4
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	4
2	1002403	THE VILLAGE OF HARLEM....NE W YORK !	78829239556	NaN	Elise	Manhattan	Harlem	4
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	4
5	1004098	Large Cozy 1 BR Apartment In Midtown East	45498551794	verified	Michelle	Manhattan	Murray Hill	4

5 rows × 24 columns

Remove Duplicates

```
df.drop_duplicates(inplace = True)

df.info()

<class 'pandas.core.frame.DataFrame'>
Index: 101410 entries, 0 to 102057
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                     101410 non-null  int64
1   NAME                                  101410 non-null  object
2   host id                               101410 non-null  int64
3   host_identity_verified                101134 non-null  object
4   host name                             101410 non-null  object
5   neighbourhood group                  101384 non-null  object
6   neighbourhood                          101394 non-null  object
7   lat                                   101402 non-null  float64
8   long                                  101402 non-null  float64
9   country                               100884 non-null  object
10  country code                          101288 non-null  object
11  instant_bookable                      101314 non-null  object
12  cancellation_policy                   101340 non-null  object
13  room type                             101410 non-null  object
14  Construction year                     101210 non-null  float64
15  price                                 101171 non-null  float64
16  service fee                           101142 non-null  float64
17  minimum nights                        101016 non-null  float64
18  number of reviews                     101228 non-null  float64
19  last review                           51867 non-null   datetime64[ns]
20  reviews per month                     101410 non-null  float64
21  review rate number                     101103 non-null  float64
22  calculated host listings count         101092 non-null  float64
23  availability                           100990 non-null  float64
dtypes: datetime64[ns](1), float64(11), int64(2), object(10)
memory usage: 19.3+ MB
```

Fixing inconsistencies

```
df['neighbourhood group'] = df['neighbourhood group'].str.title()
df['neighbourhood group'] = df['neighbourhood group'].replace({'Brooklyn':
    'Brooklyn', 'Manhatan': 'Manhattan'})
df['neighbourhood group'].value_counts()
```

	count
neighbourhood group	
Manhattan	43279
Brooklyn	41364
Queens	13120
Bronx	2678
Staten Island	943

dtype: int64

```
raw_dates = pd.read_csv('/content/Airbnb_data.csv', skipinitialspace=True)
['last review']
```

```
test_dates = pd.to_datetime(raw_dates, errors='coerce')
print("Failed dates:", (test_dates.isna() & raw_dates.notna()).sum())
print(raw_dates[test_dates.isna() & raw_dates.notna()].head(10))
```

```
d1 = pd.to_datetime(raw_dates, format='%m/%d/%Y', errors='coerce')
d2 = pd.to_datetime(raw_dates, format='%d-%m-%Y', errors='coerce')
df['last review'] = d1.fillna(d2)
```

```
df.loc[df['last review'] > pd.Timestamp.today(), 'last review'] = pd.NaT
```

```
print("Missing:", df['last review'].isna().sum())
print("Max date:", df['last review'].max())
```

Failed dates: 34335

```
3    07-05-2019
6    10-05-2017
7    10-05-2017
10   06-09-2019
14   07-05-2019
18   07-01-2019
22   07-01-2019
23   01-01-2019
24   07-02-2019
31   06-01-2019
```

Name: last review, dtype: object
Missing: 15719
Max date: 2025-06-26 00:00:00

Descriptive Statistics

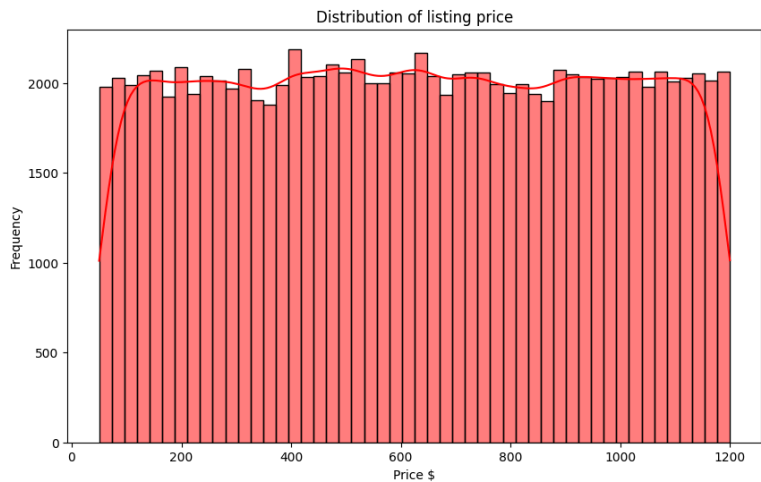
df.describe()

	id	host id	lat	long	Construction year	price	service fe
count	1.014100e+05	1.014100e+05	101402.000000	101402.000000	101210.000000	101171.000000	101142.000000
mean	2.920959e+07	4.926155e+10	40.728082	-73.949663	2012.486908	625.381008	125.043998
min	1.001254e+06	1.236005e+08	40.499790	-74.249840	2003.000000	50.000000	10.000000
25%	1.507574e+07	2.459183e+10	40.688730	-73.982570	2007.000000	340.000000	68.000000
50%	2.922911e+07	4.912069e+10	40.722300	-73.954440	2012.000000	625.000000	125.000000
75%	4.328308e+07	7.399747e+10	40.762750	-73.932340	2017.000000	913.000000	183.000000
max	5.736742e+07	9.876313e+10	40.916970	-73.705220	2022.000000	1200.000000	240.000000
std	1.626820e+07	2.853703e+10	0.055850	0.049474	5.765130	331.609111	66.313374

Visualisation

Distribution of listing prices:

```
plt.figure(figsize=[10,6])
sns.histplot(df['price'], bins = 50, kde = True, color= 'red')
plt.title('Distribution of listing price')
plt.xlabel('Price $')
plt.ylabel('Frequency')
plt.show()
```



The histogram shows a fairly even distribution of listing prices across different price ranges, indicating no particular concentration of listings in any specific price range. The kde line helps visualize this spread even more clearly, confirming that the dataset contains listings with a wide variety of prices

Distribution of different room types

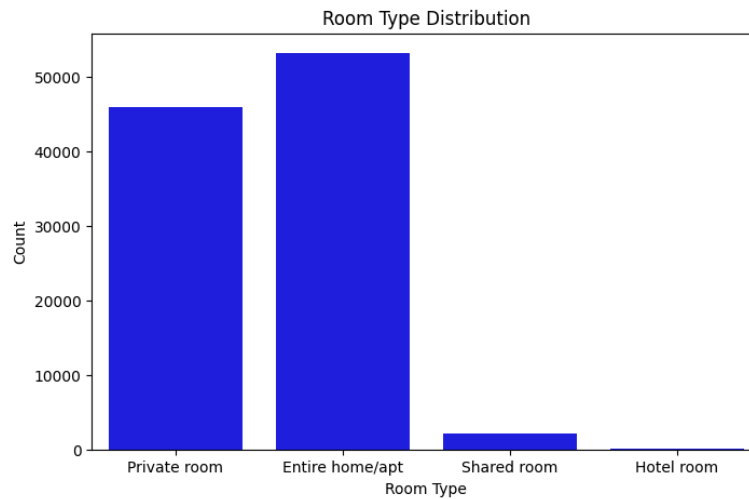
```
df['room type']
```

	room type
0	Private room
1	Entire home/apt
2	Private room
4	Entire home/apt
5	Entire home/apt
...	...
102053	Private room
102054	Private room
102055	Entire home/apt
102056	Private room
102057	Entire home/apt

101410 rows × 1 columns

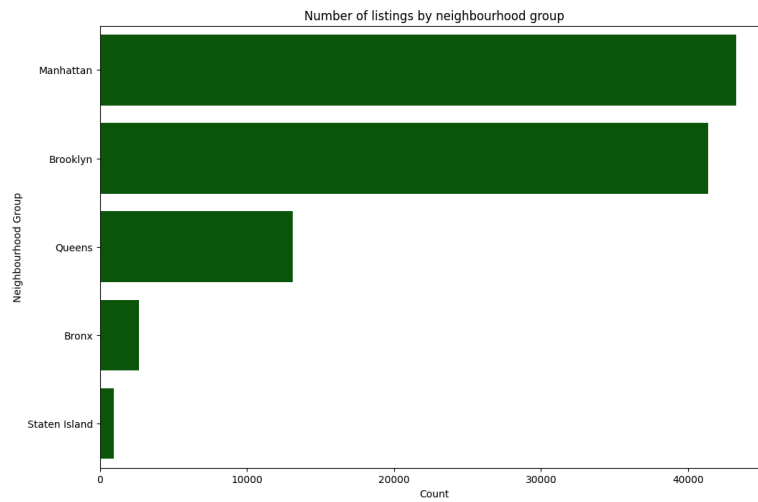
dtype: object

```
plt.figure(figsize= (8,5))
sns.countplot(x = 'room type', data = df, color = 'blue')
plt.title('Room Type Distribution')
plt.xlabel('Room Type')
plt.ylabel('Count')
plt.show()
```



Distribution of listings across different neighbourhoods

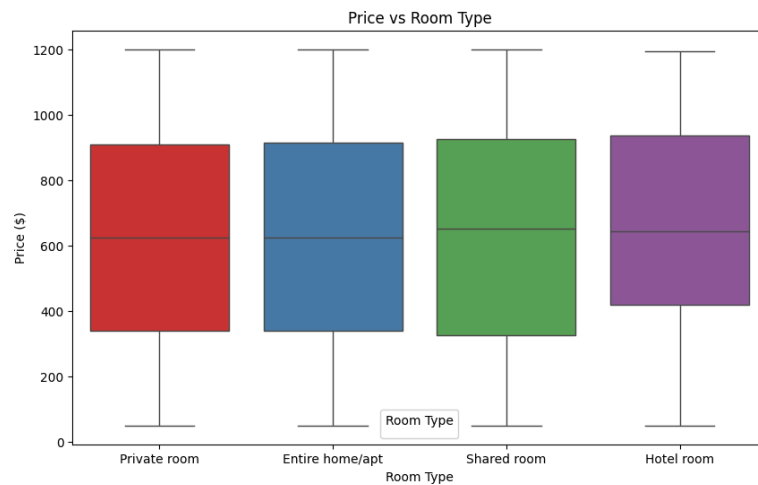
```
plt.figure(figsize = (12,8))
sns.countplot(y = 'neighbourhood group', data = df, color = 'darkgreen',
              order = df['neighbourhood group'].value_counts().index)
plt.title('Number of listings by neighbourhood group')
plt.xlabel('Count')
plt.ylabel('Neighbourhood Group')
plt.show()
```



Manhattan, Brooklyn and Queens are the neighbourhood groups where most of the properties are listed

Relationship between Price and Room Type

```
plt.figure(figsize = (10,6))
sns.boxplot(x = 'room type', y = 'price', hue = 'room type', data = df,
            palette= 'Set1')
plt.title('Price vs Room Type')
plt.xlabel('Room Type')
plt.ylabel('Price ($)')
plt.legend(title = 'Room Type')
plt.show()
```



Change of number of reviews over time

```
df['last review'] = pd.to_datetime(df['last review'])
reviews_over_time = df.groupby(df['last review'].dt.to_period('M')).size()

plt.figure(figsize = (12,6))
reviews_over_time.plot(kind = 'line', color = 'red')
plt.title('Number of Reviews over time')
plt.xlabel('Date')
plt.ylabel('Number of Reviews')
plt.show()
```

